

REAL-TIME DRONE FPV STREAMING TO VR HEADSET

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EE 430, 5G Wireless Networks

OBJECTIVE

To design and implement a 5G-enabled FPV drone system by integrating the AjnaXR SE VR Headset with the Suparna 5G Drone.



USECASES

An FPV Drone has a huge number of practical uses:

- Search and rescue operations.
- Live event broadcasting and cinematography.
- Infrastructure inspection (bridges, buildings, roads).

Why use a VR headset instead of a monitor?

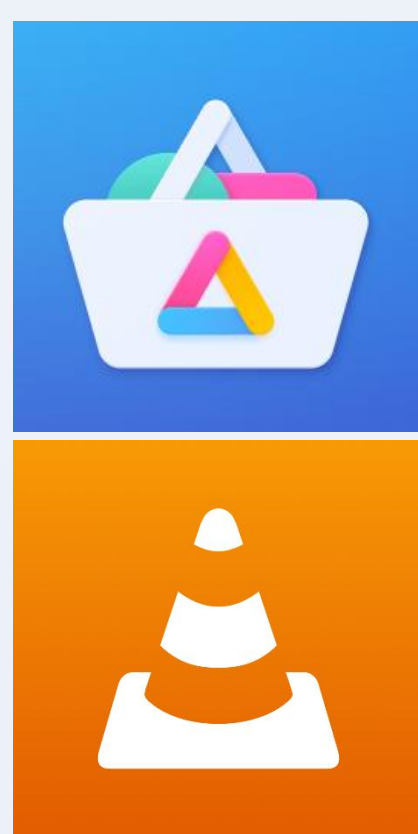
- Better vision in sunlight or brightly lit areas where a monitor can appear dim.
- Better portability and viewing experience compared to using a monitor to view drone feed.

Why use 5G and not a radio-controlled drone?

- Long range / BVLOS (beyond visual line of sight).
- High-quality video streaming.

AJNA XR VR HEADSET

- The Ajna XR SE VR Headset runs Android which allows us to install any android app on it.
- Download .apk file on laptop, transfer it to the headset and install.
- Install Aurora Store on the VR headset and install any app using it.
- Install VLC Media Player to stream any rtsp link.



SUPARNA 5G DRONE

- The drone is supposed to be controlled using iDronam software but its paid.
- Use QGroundControl as an open-source alternative.



OVERLAYING TELEMETRIES ON DRONE FEED

- Inconvenient to remove VR headset to look at important telemetries like battery, arming mode and flight mode on the computer.
- Overlay these on the video feed itself.
- Use FFmpeg to overlay and mediamtx to broadcast the updated feed.



FUTURE WORKS

- Fly the drone using the VR Headset's controllers.
- The drone currently uses a single network for both video and control. Adding a Raspberry Pi with a 5G SIM and an external camera as part of payload can enable higher-quality or specialized video (e.g., IR) and reduce control latency.
- The drone's IP will change as it connects to different 5G towers. Use a VPN to assign it a fixed virtual address.